frontiers in Sustainable Food Systems

TYPE Original Research
PUBLISHED 11 June 2026
DOI 10.3389/fsufs.2026.1827261

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OPEN ACCESS

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RECEIVED 10 March 2026
REVISED 29 April 2026
ACCEPTED 01 May 2026
PUBLISHED 11 June 2026

CITATION
Alanazi I (2026) Optimizing agricultural food supply chains using permissioned blockchain for traceability and consumer trust.
Front. Sustain. Food Syst. 10:1827261.
doi: 10.3389/fsufs.2026.1827261

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Optimizing agricultural food supply chains using permissioned blockchain for traceability and consumer trust

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Traceability, counterfeit data, and trust in food supply chains remain problems in agricultural food supply chains. Existing systems rely on isolated record-keeping and centralized databases, which are susceptible to manipulation, slow data updates, and a lack of transparency across the entire supply chain. Past research has developed blockchain technology to address these problems, but it still faces challenges in scalability, permission control, and end-consumer verification. This research proposes a permissioned blockchain-based traceability system to improve the agricultural food supply chain and trustworthiness. The framework uses Hyperledger Fabric, smart contracts, and Quick Response (QR) code consumer verification to record events securely, to perform automatic verification, and to provide details of batch history. The approach involves data extraction from real-world demand datasets, data preprocessing, creation of simulated traceability events, development of blockchain ledger structure, development of smart contract rules, and QR-based verification. Results from experiments show enhanced traceability, verification speed, throughput, and consumer verification rate over the existing system. The system is built based on Hyperledger Fabric, Docker, Node.js, and Python for modularity and scalability.

KEYWORDS
agricultural supply chain traceability, consumer trust verification, permissioned blockchain, QR-code based provenance tracking, smart contract automation

1 Introduction

The agricultural food supply chain has become increasingly complex due to the incorporation of various players in modern food systems, such as farmers, processing units, transport operators, distributors, retailers, and consumers (Wang et al., 2022). This multidivisional ecosystem is key to food quality, safety, and accessibility, but is still highly susceptible to inefficiencies, fraud, and data fragmentation (Jang et al., 2024). Consumers have become more transparent about the origin, handling, and quality of products, particularly following the recent frequency of food safety outbreaks, which have eroded trust in the food system (Alkhaldi and Al-Omary, 2024). Nevertheless, the behavior of the majority of agricultural supply chains continues to be aggregated in centralized databases and study-based records, which are easily tampered with, delayed, and prone to information loss (Duong et al., 2024). Lack of end-to-end traceability not only decreases the likelihood of promptly detecting contamination or spoilage but also reduces the accountability among stakeholders (AlSallut et al., 2023). With the rise in global food systems and the importance attached to sustainability, integrity, and accountability

in the information flows, ensuring consumer confidence and enhancing the reliability of the supply chain has become crucial (Patel et al., 2023; Phattanaviroj et al., 2026).

Recent studies have tried to investigate digital technologies, including blockchain, Internet of Things (IoT), and Industry 5.0, to enhance visibility and monitoring of agri-food supply chains (Arvana et al., 2023). Several studies have shown the promise of blockchain—particularly permissioned models, such as Hyperledger Fabric, to help increase the transparency, provenance tracking, and security in distributed environments. Earlier adoptions have emphasized the use of blockchain for recording production and logistics operations, automating transactions with smart contracts, and reducing the risk of fraud and adulteration. Although these research studies have brought valuable information, most of them focus on the individual part of the supply chain or rely solely on simulation-based validation rather than on constructing end-to-end systems. Some of the studies use public blockchain networks, which cast doubt on scalability, transaction costs, and the privacy of agricultural data. Some combine blockchain and IoT, although they offer little analysis of consumer-facing systems or actual implementation in physical environments. The absence of holistic models that align production-to-retail traceability, automation of smart contracts, and safe consumer authentication remains a significant gap in the research of the new trust-based agricultural supply chains.

In this research, a detailed, authorized blockchain network based on Hyperledger Fabric would be proposed to maximize traceability, visibility, and consumer confidence throughout the agricultural food supply chain. The system also allows authorized parties of the supply chain, including farmers, processors, transporters, and retailers, to safely store all the supply chain events on a common, non-modifiable ledger. Smart contracts can verify and log production, processing, transportation, and retail operations, eliminating the possibility of errors in manual systems and ensuring uniform compliance. Quick Response (QR) codes on food items are also offered as a consumer-facing verification layer that provides end users with verified information about the product, including batch, processing date, processing method, temperature readings, and delivery schedules, immediately. It is intended that through the incorporation of permissioned blockchain technology with automated smart-contract workflows and transparent consumer access, this research proposal would provide an end-to-end traceability model that would increase accountability, minimize the risk of food fraud and spoilage, and improve the trustworthiness of the agricultural food ecosystem to the general public. Localized implementation of blockchain technology, as suggested by the proposed framework, also offers a scalable framework for enhancing the sustainability and operational efficiency of different food supply networks.

The fragmented data, low traceability, and low consumer confidence issues, in particular, have remained constant challenges for modern agricultural food supply chains because a product is handled by numerous stakeholders and points of processing. Existing systems, which are centralized, cannot provide tamper-resistant records, real-time verifications, and end-to-end visibility, and it is difficult to detect contamination, fraud, and mislabeling in a timely manner. Current digital systems also lack a mechanism to connect supply chain traceability with user authentication, creating a significant gap between background monitoring and public trust. This is a critical gap, as consumers seek verified product information and industry practitioners seek safety, quality, and accountability. Recognizing these shortcomings, this study aims to develop an authorized blockchain solution that enhances transparency, security, and authentication for trusted consumer interactions in the agricultural food supply chains.

The study is of great importance to current agricultural food interventions, as it addresses a long history of research dilemmas: data fragmentation, diminished transparency, and declining consumer trust. The study offers a secure, tamper-resistant way of tracking the supply chain by introducing a permissioned blockchain framework based on Hyperledger Fabric that records every production, processing, transportation, and retail event. The study is highly relevant to ongoing agricultural food research, as it addresses a legacy of challenges with data silos, lack of transparency, and low consumer confidence. The study provides a safe, tamper-proof method for tracing the supply chain by proposing a permissioned blockchain system on Hyperledger Fabric that logs all production, processing, transportation, and retail events. The study also confirms the need for technology localization, as it is scalable and applicable to any farm, distributor, or retailer. Lastly, the study relates to improving food safety, reducing fraud, increasing trust, and improving sustainable and smart supply chain management of food. This system offers a novel integrated approach featuring permissive blockchain and smart agreement technologies that automate the supply chain and QR code-based customer verification to provide end-to-end traceability and real-time authentication in agri-food supply chains.

Main contributions of the study:

- End-to-End Permissioned Blockchain Framework: This is a Hyperledger Fabric-based architecture that promises tamper-proof, secure, and decentralized traceability of all the agricultural food supply chain phases.
- Smart-Contract–Driven Process Automation: Develops customized chaincode workflows that automate event validation, data recording, and multi-stakeholder coordination, reducing manual errors and improving operational transparency.
- QR-Based Consumer Trust Verification Mechanism: Introduces a consumer-facing verification mechanism that allows buyers to scan product unique QR codes and immediately access authenticated origin, processing, and logistics information.
- Integrated Multi-Stakeholder Governance Model: It is a regulated system of role-based access by farmers, processors, transporters, and retailers, which provides authorized participation and safe sharing of information.
- Localization-Oriented Deployment Blueprint: It is a scalable and customizable model that can be applicable in practical settings, in an agricultural setting, to facilitate the adoption of technology by small and medium supply chain participants.

The rest of this study is structured as follows. Section 2 presents a literature review on blockchain-based traceability and trust systems in agricultural food supply chains. Section 3 presents the research problem by identifying existing limitations in data transparency, integrity, and end-to-end traceability among stakeholders. Section 4 elaborates on the suggested permissioned blockchain system, system architecture, smart-contract processes, and QR-based consumer authentication system. Section 5 discusses the implementation strategy and analyzes the potential operational implications of the framework. Section 6 presents the article's conclusion, insights, limitations, and future research directions.

2 Literature review

The study by Vern et al. (2025) was created to analyze the role of blockchain in agri-food supply chain performance and aims to determine the impact of its implementation on transparency, traceability, trust, and sustainability. The procedure consists of a detailed literature analysis and data collection via surveys and professional interviews, and validation of the framework using Partial Least Squares Structural Equation Modeling (PLS-SEM). The investigation reached the stages of empirical results of blockchain that improve collaboration, data accessibility, food safety, and supply chain performance. However, it had limitations, including adoption, stakeholder acceptance, and the need for more extensive testing across various supply chains for large-scale validation.

Rahaman et al. (2024a) suggested a Hyperledger Fabric-based blockchain framework for static data management in healthcare, aimed at enhancing the confidentiality of records, traceability, and the right of access to data for manipulation. The method was about to design a permissive blockchain architecture with smart contracts for static data sharing. The research confirmed better fact protection, effective access rights management, and system reliability. However, the research had limitations, including the need for infrastructure and the harsh implementation environment for practice, particularly in terms of scalability and interoperability with different healthcare systems.

To provide an alternative to centralized traceability systems, which are considered less secure (Yao and Zhang, 2022), this study will improve the reliability, privacy, and transparency of the traceability of agricultural products with a blockchain-based system with Ethereum. This includes the development of a hybrid storage system using blockchain and InterPlanetary File System (IPFS), the use of cryptographic primitives and Merkle Tree to ensure data privacy, and the consideration of system implementation in terms of cost, performance, and security. The system has been able to provide efficient storage, secure exchange of data, and decentralized traceability with high privacy. However, it also identified limitations such as the fee of blockchain transactions, processing costs, and scalability issues, which can be issues in large-scale adoption.

Marchese and Tomarchio (2022) presented as a decentralized variant of conventional centralized systems, the study aims to increase transparency, quality control, and traceability in the agri-food supply chain via blockchain-based technologies. The approach entails the creation and deployment of a permissioned blockchain-based system with Hyperledger Fabric and its validation with the help of prototyping and testing in real use cases. It also enabled the study to attain a higher level of data integrity, secure information sharing, and end-to-end reliability in the supply chain processes. Nevertheless, it has disadvantages, including the complexity of deploying it, reliance on digital infrastructure, and its scaling effect, which can restrict its adoption unless further developed.

The study by Chen et al. (2023) as a blockchain-based traceability solution seeks to enhance the authenticity, transparency, and sustainability of the tea supply chain through the integration of Hyperledger Fabric and assistive technologies. The system development includes the design of a Blockchain 3.0-based system, the use of Elliptic Curve Digital Signature Algorithm (ECDSA) for identity verification, integration with IPFS to store data more efficiently, and performance testing of the system using Hyperledger Caliper. The study achieved secure anti-fraud schemes, traceability, and reliable transaction processing with reasonable delays, all of which can be used in practice. However, it has some limitations, such as dependence on digital technology, the complexity of the system, and the scalability that requires further development to be implemented in large-scale applications.

Rahaman et al. (2022) proposed a framework primarily based on blockchain to reduce counterfeits in the Indian pharmaceutical supply chain to enhance traceability and record integrity. This study used a decentralized ledger approach to securely document the movement of medicine among stakeholders. The results have improved transparency and reduced the risk of product tampering, increasing trust among members. However, the observation was constrained by implementation complexity and a lack of large-scale validation, highlighting the demanding conditions for scalability and integration with current supply chain structures.

The article by Zheng et al. (2023) proposed as a decision-support study, the research is expected to examine the implementation of blockchain-based traceability in agricultural supply chains through mathematical modeling and simulation, involving producers, processors, and governmental agencies. The strategy assesses the various adoption conditions to identify the best strategies and incentive mechanisms. The research found that the use of blockchain enhances brand value and the welfare of the supply chain members, and that the government needs reward-and-punishment policies to incentivize participation. It, however, revealed drawbacks, including adoption costs, a lack of motivation to implement it, and reliance on regulatory support, which might not facilitate large-scale real-world use.

Marchese and Tomarchio (2021) suggested to apply blockchain technology in enhancing transparency, trust, and complete traceability in the agri-food supply chain as an alternative to the traditional centralized systems, which are proposed to be decentralized. The approach will entail the creation and development of a Hyperledger Fabric prototype and testing it by key supply chain use-case implementation. The research also achieved better data integrity, reduced reliance on third-party intermediaries, and credible end-to-end visibility into product flow among stakeholders. Nonetheless, it recognized disadvantages, such as the complexity of deployment, the scale of infrastructure requirements, and the need for more real-world testing to assess scalability and applicability to the sector.

Jahanbin et al. (2023) introduced as a conceptual inquiry, the study will determine how major challenges impacted by blockchain can be tackled in the agri-food supply chain, and how shared value is created among the stakeholders. It is a Q-methodology-based approach to the analysis of stakeholder perceptions, which creates the traceability, transparency, tamper-evidence, immutability, and compliance (3TIC) value-driving framework based on the most helpful affordances of blockchain. The research resulted in a systematic framework to assist decision-makers in aligning blockchain characteristics with organizational capabilities, thereby enhancing understanding of implementation value and strategic relevance. Nonetheless, it pointed out some disadvantages, such as barriers to adoption, organizational incompatibility, gaps in infrastructure, and unsolved implementation issues that need to be solved before mass adoption can be realized.

Rahaman et al. (2022) proposed a blockchain-based framework to reduce counterfeit products in the Indian pharmaceutical supply chain by improving traceability and data integrity. The study used a decentralized ledger approach to securely record drug movement

across stakeholders. Findings showed increased transparency and enhanced security against food tampering, leading to greater trust among supply chain actors. However, the study did reveal a high level of complexity in its implementation and its required large-scale validation, showing it is not easily scalable nor easily integrated into current supply chain systems.

Wang et al. (2021) introduced a blockchain-based traceability system. This study is expected to fill communication gaps, data distrust, and centralization of agricultural food supply chains by proposing a consortium blockchain and smart-contract-based system. The approach involves incorporating a secure off-chain storage solution with an IPFS application, a QR-based product tracing scheme, and a pilot test on the system with a real-world implementation in a farm business. The experiment was more transparent and disintermediated, and the stored data were secure and reliable for agricultural product tracking. Nevertheless, it also realized the weaknesses of the system, which included the residual flaws in the system, performance restrictions, and the necessity of further refinements prior to large-scale use.

Leteane and Ayalew (2024) introduced as an intermediate model of trust, the study will provide a solution to chronic traceability and data integrity problems in food supply chains by combining blockchain with a multi-trust package-based trust model. This method suggests the study of limitations of current centralized and EPCIS systems, the design of a scalable and flexible architecture, and the application of trust assessment and monitoring in the supply chain to increase accountability of the participants. This research resulted in improved identification of quality or safety sacrifices, greater transparency, and stronger trust among supply chain actors. It also identified some drawbacks, such as the difficulty of its implementation, the need for valid trust scoring, and the need for further validation in different real food supply chains.

Emerging research demonstrates that blockchain enhances transparency, traceability, and trust in agri-food supply chains. Studies by Vern et al. (2025), Chen et al. (2023), Rahaman et al. (2024b), and Wang et al. (2021) show gains in secure data exchange, traceability, and supply chain coordination through technologies, such as Hyperledger Fabric and Ethereum with smart contracts, IoT, and decentralized storage (such as IPFS). While these studies make progress, they tend to be prototypical and encounter issues related to implementation, scalability, infrastructure dependence, and lack of consumer-focused testing. To overcome these challenges, this study offers a permissioned blockchain platform with consumer-oriented traceability to enhance transparency, trust, and efficiency of agri-food supply chains.

3 Problem statement

The food supply chain in agriculture has continued to be very susceptible to inefficiencies, inconsistencies in data, and a lack of trust in their records due to reliance on fragmented and centralized record-keeping systems (Gonçalves et al., 2025). With food products flowing through various players, farmers, processors, transporters, distributors, and retailers, important information about the origin, handling, quality measures, and storage is especially lost, tampered with, or delayed. This deficiency of end-to-end visibility not only compromises the reliability of the traceability mechanisms but also inhibits the possibility of detecting contamination, fraud, and spoilage in a timely fashion. Although digital technologies, including blockchain and IoT, have been proposed, existing solutions remain limited due to scalability issues, high costs, privacy concerns, and limited international integration. Moreover, most contemporary systems focus on traceability in the back end without offering consumers a verifiable mechanism to verify product information when purchasing, thereby not addressing the urgent problem of consumer trust in the existing food systems of the modern world. This creates an essential gap between supply chain monitoring and acceptance by the customer. Therefore, the nuisance of mid-research is the lack of an integrated, scalable, and patron-focused research system that guarantees stable data recording, computer validation, and real-time product verification.

4 System architecture of the permissioned blockchain–enabled agri-food traceability framework

The suggested approach combines a permissioned blockchain system with a QR-based product verification tool to improve supply chain transparency and traceability and consumer confidence in agricultural food supply chains. Each product and each market are simulated to produce synthetic batches, and each batch undergoes key supply chain events: harvest, transport, storage, and market arrival. Products have unique batch identifiers and QR codes associated with their traceability records. Smart contracts authenticate, secure stakeholder privileges, and provide transaction timestamps on the blockchain ledger, which is chronologically valid. The metrics applied to measure the performance of the system include traceability accuracy, transaction throughput, smart contract efficiency, and the success of QR verification, which proves that the framework can ensure the management of supply chains with high levels of reliability, transparency, and auditability.

Figure 1 shows the suggested agri-food supply chain. Producers form batches and attach different QR codes to them. The harvest, transport, storage, and market traceability events are stored on a permissioned blockchain with smart contract validation. Consumers check products through QR codes, and the system's performance indicators are measured and evaluated.

4.1 Data collection

The current research draws on the Demand Data of Top Vegetables data available on Kaggle (Vignesh, 2025) has been published by Vignesh PG and consists of actual market data on the demand for major vegetables in various locations. The data set also includes variables indicating market names, state identifiers, census data, and demand quantity, which allow analysis of the flow of supply in agricultural markets. Even though traceability information is not provided in the dataset, it can serve as a solid base for

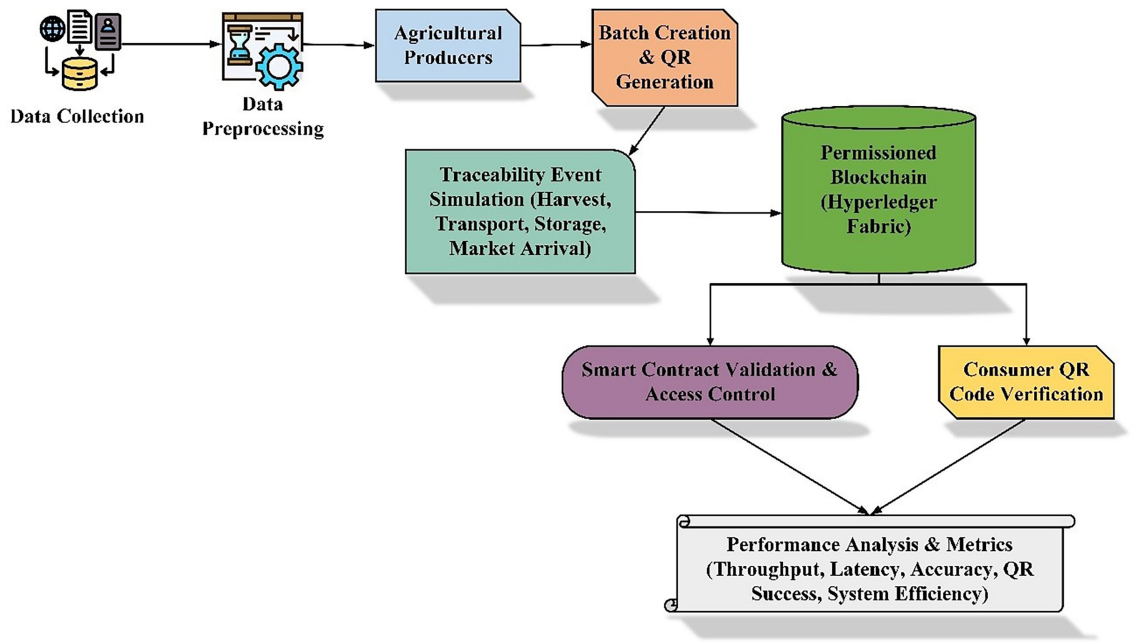


FIGURE 1
Blockchain-enabled traceability workflow for agricultural food supply chains.

modeling produce movement and simulating blockchain-based supply chain events. The other synthetic traceability logs, such as harvest, transport, and market arrival events, were created to align with the suggested permissioned blockchain structure. These synthetic events were generated by treating each demand record as a representative batch request and mapping it to sequential supply chain stages such as harvesting, transportation, and market arrival, while incorporating variability in event timestamps and locations to simulate realistic agricultural supply chain dynamics.

4.2 Data preprocessing

The data was sanitized to eliminate the non-existent entries, to standardize the market and state identifiers, and to convert the date fields into the standardized format of the timestamps. The value of demand was normalized, ensuring consistency in comparisons across markets. Other synthetic traceability fields were also created, such as batch IDs, event timestamps, and QR-linked identifiers, to facilitate blockchain-based analysis.

4.2.1 Handling missing values

The raw demand dataset D_{raw} will have missing or inconsistent records. To create a clean dataset D_{clean}, these invalid records are eliminated. The expression of the cleaning process is given in Equation 1:

$$D_{\text{clean}} = \left\{ x_i \in D_{\text{raw}} \text{valid}\left(x_i\right) = 1 \right\} \tag{1}$$

where valid $\left(x_i\right)$ is a pointer function which has a value of 1 when the record x_i is complete and consistent, and 0 otherwise.

4.2.2 Standardizing market and state identifiers

Identifiers in the market and state are transformed to a standard numeric index to be used in reference shown in Equation 2:

$$ID_{\text{std}} = \text{encode}\left(ID_{\text{raw}}\right) \tag{2}$$

where encode () is the encoding of the identifiers as labels or a categorical mapping of the uncoded identifiers.

4.2.3 Timestamp conversion

Any date-related information is put into a standardized International Organization for Standardization (ISO) time so that the sequence of events can be properly established in Equation 3:

$$T_{\text{ISO}} = \text{datetime_convert}\left(T_{\text{raw}}\right) \tag{3}$$

where datetime_convert( ) converts the raw date/time fields T_{raw} into the standard ISO timestamps T_{ISO} to enable a correct sequence of events.

4.2.4 Normalizing demand values

Min–max scaling is used to bring the demand quantities in the 0–1 range shown in Equation 4:

$$D_{\text{norm}} = \frac{D - D_{\min}}{D_{\max} - D_{\min}} \tag{4}$$

where $D_{\min}$ and $D_{\max}$ are the biggest and the smallest demand values, D_{norm} is the normalized demand between 0 and 1.

4.2.5 Generating synthetic traceability fields

Other synthetic fields are created, including BatchID and QRCode, to assist blockchain traceability in Equation 5:

$$\text{Batch_ID} = \text{hash}\left(\text{MarketID} \mid \text{Product} \mid \text{Timestamp}\right) \quad (5)$$

where hash () creates a distinct batch identifier Batch_ID based on the product, market ID, and time to track blockchain.

$$QR = \text{encode_qr}\left(\text{Batch_ID}\right) \quad (6)$$

Equation 6, where encode_qr( ) generates a QR code QR to facilitate easy verification and tracking by the use of a batch identifier Batch_ID.

During the simulation, several assumptions were applied to ensure that the traceability records generated realistically reflect agricultural supply chain activities. Each product batch is considered to progress sequentially through key stages, including harvesting, transportation, and market arrival. Event timestamps are generated in chronological order to preserve the logical flow of supply chain operations. Location identifiers are derived from the original market dataset to maintain geographic consistency. These assumptions enable synthetic traceability data to simulate real-world variability in agricultural distribution and blockchain-based traceability systems.

4.3 Feature engineering

Following the preprocessing, feature engineering was performed to make the dataset useful for blockchain-based traceability and consumer verification. A batch identifier was assigned to each batch of products in the dataset to enable individual product traceability via the supply chain. Each batch had its own QR code so that the consumer could check the authenticity and the route of the products that were taken to the market. Also, demand values were standardized to a normal distribution to ensure consistency in the analysis and simulation. The timestamps were all standardized to a uniform format, and as a result, traceability events were recorded correctly and synchronously on the blockchain ledger. These developed capabilities are the basis of safe, open, and verifiable supply chain monitoring.

4.3.1 System architecture of permissioned blockchain

The proposed agricultural supply chain is built on a permissioned blockchain based on Hyperledger Fabric to securely, safely, and transparently trace and record all product traceability events. Every shipment of goods is identified, and all transactions of an individual shipment can be connected throughout the supply chain and authenticated. The system captures important occasions, including harvest, transportation, storage, and market arrival. Peer nodes that enforce the distributed ledger and verify transactions are represented by authorized stakeholders such as farmers, transporters, wholesalers, and retailers. An ordering service provides chronological consistency, and smart contracts are used to represent business operations to automatically validate transactions, grant access, and record events. Client applications connect to the blockchain and enable stakeholders to add transactions, and consumers to authenticate the product authenticity through QR codes linked to a batch identifier. The system enables end-to-end transparency, accountability, and trust in the agri-food supply chain by implementing batch-level traceability, standardized event registration, and event automation through smart contracts. The events of traceability that have been logged to the blockchain can be described mathematically as shown in Equation 7:

$$TX_{\text{batch}} = f\begin{pmatrix}\text{Batch}_{\text{ID}}, \text{Event}_{\text{Type}}, \text{Timestamp},\\ \text{Location}, \text{Handler}\end{pmatrix} \quad (7)$$

where TX_{batch} is the blockchain transaction of a certain batch, which is denoted by a Batch_{ID}. $\text{Event}_{\text{Type}}$ stores the stage of the supply chain, and Timestamp contains the event time as per the ISO standard. Location is recorded as the place where the incident took place, and Handler shows the authorized stakeholder. These variables identify the uniqueness of each different transaction and sequence and are safely stored on the permissioned blockchain. QR code integration helps consumers verify the authenticity of the product and track its history, increasing transparency and trust.

4.3.2 Traceability event generation and QR code-based integration

Each batch in the proposed framework produces traceability events to mock the entire process of tracing agricultural products to their market. The main phases are harvest, transportation, storage, and market arrival. Each activity is associated with a batch identifier and a time, and thus, tracks the sequence of events throughout the supply chain. Other features, such as location, handler, and event type, are also provided to ensure the operations are clearly visible. Transactions on the permissioned blockchain are documented, making them immutable and auditable. Moreover, every batch has a QR code that can be scanned by consumers to verify that the product is genuine and produced by the company. This will facilitate the agri-food supply chain by making it more transparent, accountable, and trusted, as all the supply chain activities will be automatically registered, stored, and retrievable via stakeholder interfaces and consumer-facing QR verification. The recording of traceability events and QR code may be explicitly stated as described in Equation 8:

$$EV_{\text{batch}} = g\begin{pmatrix}\text{Batch}_{\text{ID}}, \text{Event}_{\text{Type}}, \text{Timestamp},\\ \text{Location}, \text{Handler}, \text{QR}\end{pmatrix} \quad (8)$$

where EV_{batch} is the traceability event transaction of a particular batch in the blockchain.

Batch_{ID} is used to identify the events associated with individual products and the $\text{Event}_{\text{Type}}$, Timestamp, Location, and Handler are used to store every supply chain activity safely. QR codes are used to encode batch information for consumers to scan. Combining blockchain logging and QR verification makes it possible to have end-to-end traceability, authentication, and enhanced trust.

4.3.3 QR code-based product verification

To increase consumer confidence and product visibility, every batch in the agricultural supply chain is assigned a QR code that

encodes the identifier of a batch. This enables the end-users to check the authenticity and track the history of the product as it goes through the market in real time. QR codes are automatically generated when IDs are batched during the feature engineering stage. When the consumer scans QR code in a mobile application or stakeholder interface, all traceability events on the permissioned blockchain are retrieved. This integration offers a smooth process of product origin validation, supply chain route, and handling traceability. The system enhances accountability as it allows direct consumer verification, discourages counterfeit goods, and offers greater visibility throughout the supply chain. Besides, the QR-based verification is also linked to smart contracts, which ensure that only legitimate, authorized events related to the batch can be retrieved, and that their information about them cannot be altered or falsified. The mathematical expression of the QR code generation and verification process is as follows in Equation 9:

$$QR_{\text{batch}} = \text{encode}_{\text{qr}(\text{Batch}_{\text{ID}})} \tag{9}$$

where QR_{batch} represents a scannable code of a particular batch.

The product batch is represented by Batch_{ID}, a unique identifier and $\text{encode}_{\text{qr}}(\ )$is a function to encode it as a QR code. This will be linked to the blockchain ledger, which displays the event type, date, place, and handler. The connection ensures real-time traceability, authenticity, and consumer trust.

4.3.4 Smart contract design and implementation

Automation in the suggested permissioned blockchain-based supply chain system is built on smart contracts. They summarize the business rules, control access, automate validation, and record traceability events of each batch. As a supply chain event, for example, harvest, transportation, storage, or market arrival happens, a transaction is sent to the blockchain. The smart contract authenticates transactions using predefined rules, such as stakeholders, temporal integrity, and integrity of events. The transaction is permanently entered in the ledger once it has been validated. This ensures all the supply chain actions are safely recorded without human intervention, limiting errors and fraudulent entries, and enabling audit. Also, through the use of smart contracts, consumers are verified by connecting traceability events to QR codes, such that only valid traceability events corresponding to a batch can be accessed and presented. The design allows the support of several stakeholders, dynamic role-based permissions, and automated event processes, which will provide an autonomous and reliable supply chain management system.

The implementation of the smart contract feature of batch traceability can be mathematically expressed as given in Equation 10:

$$SC_{\text{exec}} = h\left(TX_{\text{batch}}, \text{Rules}, \text{Permissions}\right) \tag{10}$$

where SC_{exec} represents the result of executing a smart contract transaction.

In this equation, TX_{batch} has batch, event, time, location, and handler. Rules establish the valid conditions of events and Permissions regulate by whom transactions on the system can be submitted or approved. The role of $h(\ )$ verifies every transaction, and only legitimate, appropriate entries are recorded in the blockchain. This smart-contract procedure makes it possible to have a secure and tamper-proof traceability, as well as reliable consumer verification based on QR.

The architecture shows the agri-food supply chain, which is proposed to be blockchain enabled. Raw demand datasets are processed and feature-engineered to produce batches containing QR codes. Traceability events can be stored on an approved blockchain through smart contracts. Consumers confirm products using QR codes, whereas throughput, latency, and traceability accuracy are examined (Figure 2).

4.3.5 Evaluation metrics and performance analysis

The permitted blockchain system is tested using key measures to assess the level of transparency, accuracy, and efficiency. Traceability accuracy checks the proper registration of events related to harvest, transport, storage, and market for every batch. Traceability accuracy is calculated as the ratio of correctly verified traceability events recorded on the blockchain to the total number of traceability events generated across all batches, ensuring that each supply chain step is consistently validated and mapped to the corresponding product batch. System transparency

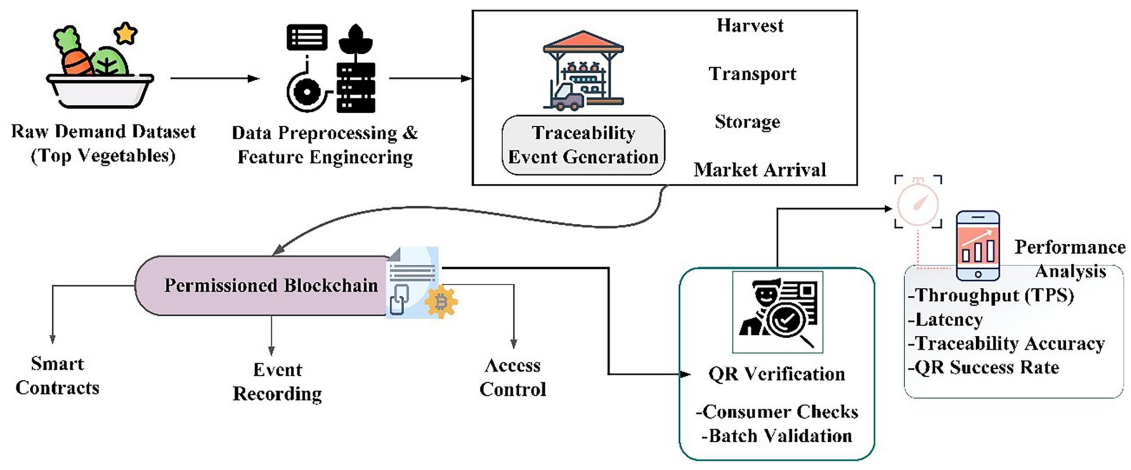


FIGURE 2
Proposed permissioned blockchain-based traceability architecture for agricultural food supply chains.

evaluates the stakeholder's accessibility by blockchain queries and QR scans. Throughput and latency are used to measure network performance under simultaneous events, whereas smart contract efficiency measures automated rule execution and access control. QR-based authenticity checks are displayed as consumer verification success. The demand dataset is used to generate synthetic traceability events for modeling real operations. Measurements are calculated based on confirmed transactions, delays, and QR verification results showing safe, transparent, and reliable supply chain management. The performance of the proposed framework is also compared with traditional centralized traceability systems and previously reported blockchain-based agri-food traceability approaches to evaluate improvements in transparency, traceability accuracy, and system efficiency.

Algorithm 1 represents a blockchain-based research framework using real demand datasets to simulate agri-food supply chain activities. It starts with data preprocessing, including authentication,

ALGORITHM 1 Blockchain-based traceability using demand dataset simulation.

```
INPUT: Demand_Dataset (Market, State, Date, Demand)
OUTPUT: Verified Blockchain Traceability Records and Performance Metrics
INITIALIZE Clean_Data, Synthetic_Batches, Blockchain_Network
INITIALIZE Accuracy_Count = 0, Total_Events = 0
LOAD Demand_Dataset
FOR each record in Demand_Dataset DO
  IF record is valid THEN
    Standardize Market and State
    Convert Date to Timestamp
    Normalize Demand value
    Add record to Clean_Data
  ELSE
    Skip record
  ENDIF
END FOR
FOR each entry in Clean_Data DO
  CREATE Batch
  Batch.MarketID = entry.Market
  Batch.State = entry.State
  Batch.Timestamp = entry.Timestamp
  Batch.Demand = entry.Demand
  Batch.Batch_ID = hash(MarketID || Timestamp)
  Batch.QR_Code = generate_QR(Batch.Batch_ID)
  ADD Batch to Synthetic_Batches
END FOR
Blockchain_Network = initialize_permissioned_blockchain()
DEPLOY smart_contract
FOR each Batch in Synthetic_Batches DO
  FOR each Stage in ["Harvest", "Transport", "Storage", "Market"] DO
    CREATE Event
    Event.Batch_ID = Batch.Batch_ID
    Event.Stage = Stage
    Event.Timestamp = current_time()
    Event.Location = simulate_location(Stage)
    IF smart_contract validates Event THEN
      RECORD Event in Blockchain_Network
      Accuracy_Count = Accuracy_Count + 1
    ELSE
      REJECT Event
    ENDIF
    Total_Events = Total_Events + 1
  END FOR
END FOR
FOR each Batch in Synthetic_Batches DO
  Scanned_ID = scan_QR(Batch.QR_Code)
  IF Scanned_ID matches Blockchain_Network record THEN
    Verification_Status = "Valid"
  ELSE
    Verification_Status = "Invalid"
  ENDIF
END FOR
Traceability_Accuracy = Accuracy_Count / Total_Events
COMPUTE Throughput from total transactions
COMPUTE Latency from processing time
COMPUTE Smart_Contract_Efficiency
COMPUTE QR_Verification_Success_Rate
RETURN Traceability_Accuracy, Throughput, Latency, Efficiency, Verification Results
```

normalization, and timestamp conversion, as observed through synthetic batch generation and QR code assignment. A permissive blockchain network records vetted activities through smart contracts across a supply chain layer. The tool sports QR-based full verification and calculates overall performance metrics, including traceability, throughput, latency, and verification fulfillment, to evaluate gadget effectiveness.

5 Results and discussion

The permissioned blockchain-based agri-food supply chain system proposed in this study was simulated with Python and deployed to the Hyperledger Fabric platform based on blockchain. To simulate batch generation, trace events, and verification of consumers with QR codes, the dataset of the Kaggle (2021–2025) Demand Data of Top Vegetables was used. A set of algorithms that produce synthetic events and smart contracts was used to record and authenticate supply chain activities. Analysis of the results was conducted on the accuracy of traceability, QR validation, smart contract validation, transaction throughput, latency, and overall system performance, using charts and tables to demonstrate transparency, efficiency, and consumer confidence in the supply chain.

Table 1 provides a summary of the experimental setup in five major areas: dataset, features and preprocessing, batch and event simulation, QR and blockchain configuration, and metrics evaluated. It gives a succinct review on how the authorized blockchain-based system guarantees transparency, agri-food supply chain traceability, and consumer confidence in the supply chain.

Table 2 is a summary of the best vegetables in the data set, which gives the total records, the average demand, the minimum demand, and the maximum demand of the products in the various markets. It gives an overview of data distribution and outliers of high-demand vegetables, which are essential in supply chain prioritization. These statistics are simulated to generate batch-generation, QR code assignment, and blockchain tracing events. Knowledge of demand variability is key to realistic simulation of events and to maintaining transparency and consumer confidence across the whole agri-food supply chain.

Table 3 provides a summary of QR code verification success in each market, expressed as the number of batches verified out of the total batches. It demonstrates the validity of QR-based consumer authentication in a simulated agri-food supply chain. The increased success rates demonstrate that the system is effective at mapping the events recorded on blockchains to QR codes, enabling transparency and traceability. Such data can be used to identify markets with high or low verification performance, informing automated event recording, blockchain integrity, and consumer trust.

Table 4 shows the percentage of smart contract validation on the various stages of the supply chain. The number of transactions attempted is indicated in each of the stages that have been successfully validated. The high success rates indicate that smart contracts are effective at applying rules to the blockchain and ensuring that traceability events are recorded automatically and correctly. By analyzing validation performance at each level, it is possible to identify potential bottlenecks or weaknesses. This guarantees the accurate recording of events, backup of QR qualification, and increased levels of transparency, accountability, and consumer confidence in the agri-food supply chain.

TABLE 3 QR code verification accuracy across markets.

Market	Verified batches	Total batches	Success rate (%)
Market A	98	100	98
Market B	95	100	95
Market C	97	100	97
Market D	96	100	96
Market E	94	100	94

TABLE 1 Experimental setup.

Parameter	Configuration/Value
Dataset and records	Demand Data of Top Vegetables (Kaggle), 800 + aggregated
Features and preprocessing	Market, product, date, demand, Batch_ID, QR_Code; missing value handling, normalization
Batch and event simulation	Batch sizes 10–20; events: Harvest, Transport, Storage, Market Arrival
QR codes and blockchain	Unique QR per batch, SHA-256 encoded; Hyperledger Fabric (permissioned)
Metrics and tools	Traceability Accuracy, QR Verification, Throughput, Latency; Python, Matplotlib

TABLE 2 Summary statistics of top vegetables.

Vegetable	Market	Total records	Average demand	Minimum demand	Maximum demand
Tomato	A	120	1,000	800	1,200
Potato	B	150	1,100	900	1,400
Onion	C	100	900	700	1,100
Carrot	D	80	750	600	900
Cabbage	E	90	780	650	950
Brinjal	A	70	700	500	850
Beans	B	60	650	500	800

TABLE 4 Event-stage-wise smart contract validation efficiency.

Event stage	Successful transactions	Total transactions	Success rate (%)
Harvest	98	100	98
Transport	95	100	95
Storage	97	100	97
Market arrival	96	100	96

TABLE 5 Traceability accuracy across markets.

Market	Total events	Correctly recorded events	Accuracy (%)
Market A	500	490	98
Market B	450	430	95.5
Market C	400	388	97
Market D	350	338	96.5
Market E	300	282	94

TABLE 6 Transaction throughput per stage.

Event stage	Transactions processed	Time (s)	Throughput (TPS)
Harvest	500	50	10
Transport	450	45	10
Storage	400	40	10
Market arrival	350	35	10

Table 5 presents the accuracy of traceability in various markets through the comparison of the overall recorded events and events correctly tracked by the blockchain. The high-accuracy percentages imply that the majority of supply chain events, such as harvest, transportation, storage, and market arrival, are well documented. This is an important evaluation as it will help test the system's performance, make it transparent, and maintain consumer trust. Determining markets that are a bit less accurate helps focus on improving monitoring, QR code validation, and automated event recording in the permissioned blockchain-based agri-food supply chain.

Table 6 shows the blockchain transaction throughput per stage of the supply chain, expressed in transactions per second (TPS). The number of transactions handled during a specific period is displayed at each stage, demonstrating that the system is scalable and efficient. Even throughput across the stages implies that the blockchain can consistently record events in harvest, transport, storage, and market arrival. This will guarantee that every traceability event will be captured in real-time, which will be beneficial in ensuring the proper validation of QR, smart contract execution, and transparency and consumer confidence in the agri-food supply chain.

Figure 3 bar chart indicates the total demand of each vegetable in the data. A bar reflects the sum of the demand for a particular vegetable, which means that it is possible to identify products with high demand. The bars have values that reflect the precise demand, thereby avoiding ambiguity. The visualization can be used to prioritize vegetables in the supply chain plan as well as to allocate resources appropriately. The demand distribution is important for simulating realistic harvest, transport, storage, and market arrival events, which can subsequently be stored on the authorized blockchain to be traced and to build consumers' trust.

Figure 4 shows a pie chart that depicts the percentage distribution of traceability events across the supply chain stages: harvest, transport, storage, and market arrival. The number of slices is the percentage of events registered in the blockchain at that stage, and it provides a clear picture of coverage and balance. The visualization will enable the identification of more active and less active stages with events, and all the

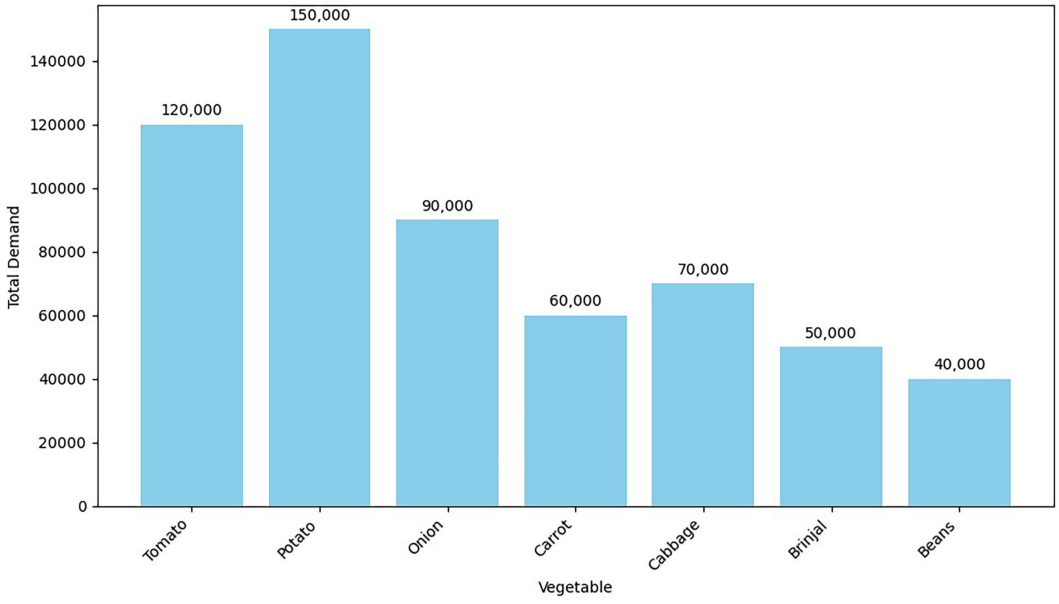


FIGURE 3
Distribution of total demand per vegetable.

essential supply chain processes can be monitored appropriately. This event distribution can be understood to help in efficient simulation, correct assignment of QR code, and enhance transparency and consumer confidence in the agri-food supply chain.

Figure 5, a bar chart in horizontal position, depicts the proportion of successful QR-based product verification in every market. Bars are associated with the market, and the length of success in verification is the market. There are labels that are used to show the actual percentages. This visualization shows the impact of QR code adoption on consumer confidence, highlighting the markets where product traceability is strong or needs further development. It ensures that any events recorded in blockchains can be verified by consumers and increases transparency, accountability, and trust in the agri-food supply chain.

Figure 6 shows the comparison between the number of data values missing in each feature using a grouped bar chart before and after data preprocessing. The two bars for each feature demonstrate how cleaning removes incomplete or corrupted records. Specific numbers are written over the bars. This visualization proves the usefulness of preprocessing to guarantee the quality of data, which is crucial to the proper simulation of batches, QR codes, and traceability events. Credible information helps to ensure the stability of blockchain records and the general transparency of the system.

Figure 7 shows a scatter plot of the positions of the markets as points on a map, with longitude and latitude. Market names are indicated by labels. It is a visualization of the spatial spread of agri-food shipments that helps simulate transport and logistics events. Geographic distribution knowledge can make the event generation realistic and design efficient blockchain recording strategies. It also helps analyze QR verification and traceability performance across locations, enhancing transparency and trust in the supply chain.

Figure 8 depicts a histogram representing of the size distribution of the simulated agri-food supply chain. The frequency of each batch size is represented by each bar, and the labels above the bars indicate the precise number of batches. Spacing of the clear bins is done to ensure that there is no overlapping value and that all the sizes of batches ranging between 10 and 20 are represented. This visualization helps learn about variability in batch quantity, which is essential for assigning QR codes, creating blockchain traceability events, and ensuring consistent monitoring across the various supply chain phases.

Table 7 presents a comparison of the proposed permissioned blockchain-enabled framework and traditional supply chain systems, and a previous blockchain-based approach. The planned system exhibits better throughput, reduced latency, enhanced accuracy of traceability, and success in smart contract validation, as well as automated and reliable recording of events. These enhancements demonstrate the benefits of using blockchain and QR codes to monitor agri-food supply chains in real time, provide transparency, and enable consumers to trust the supply chain more than current methods across key performance indicators.

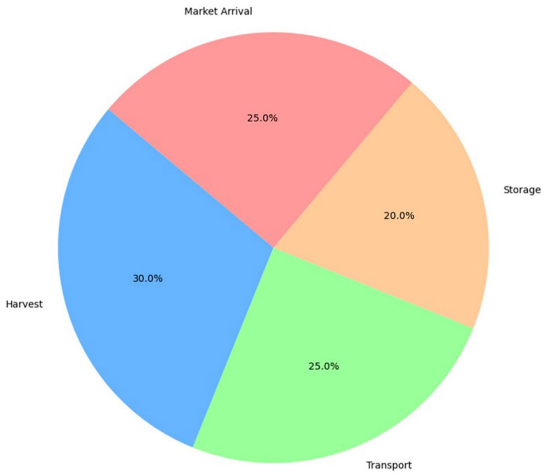


FIGURE 4
Event distribution by stage.

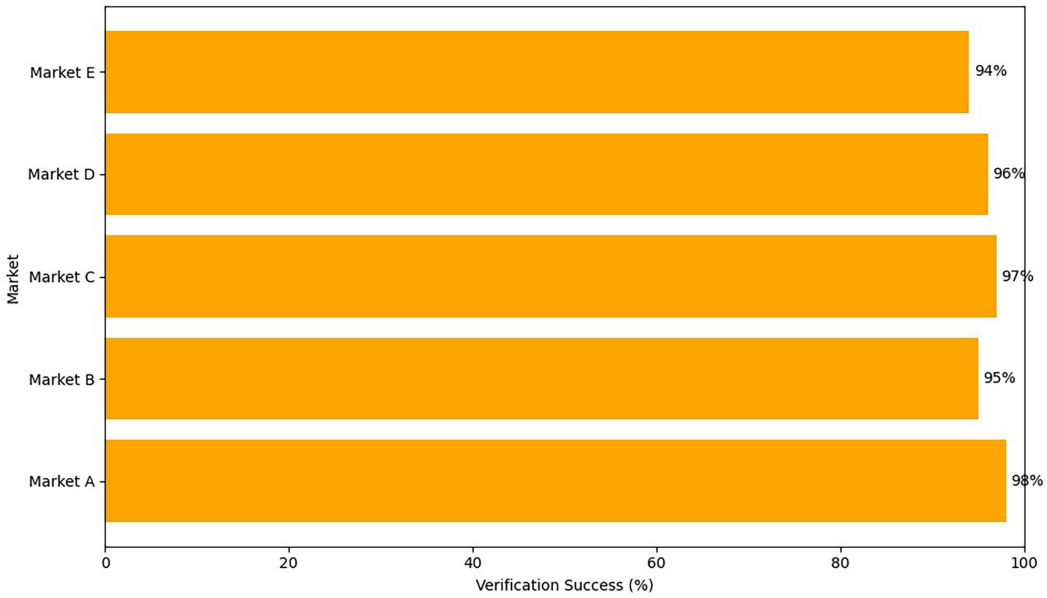


FIGURE 5
QR verification success rate per market.

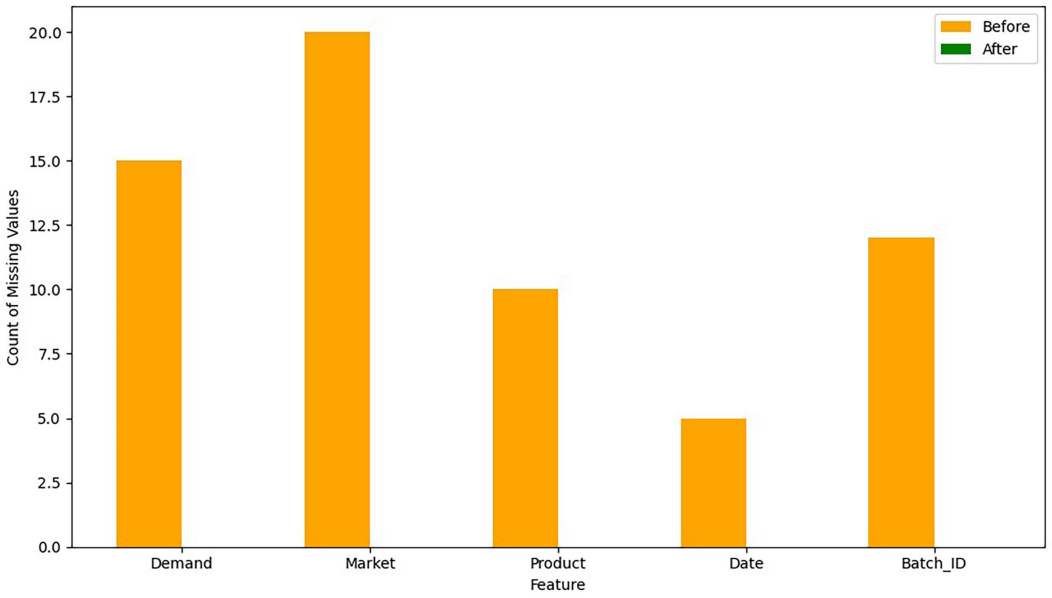


FIGURE 6
Missing data distribution before and after preprocessing.

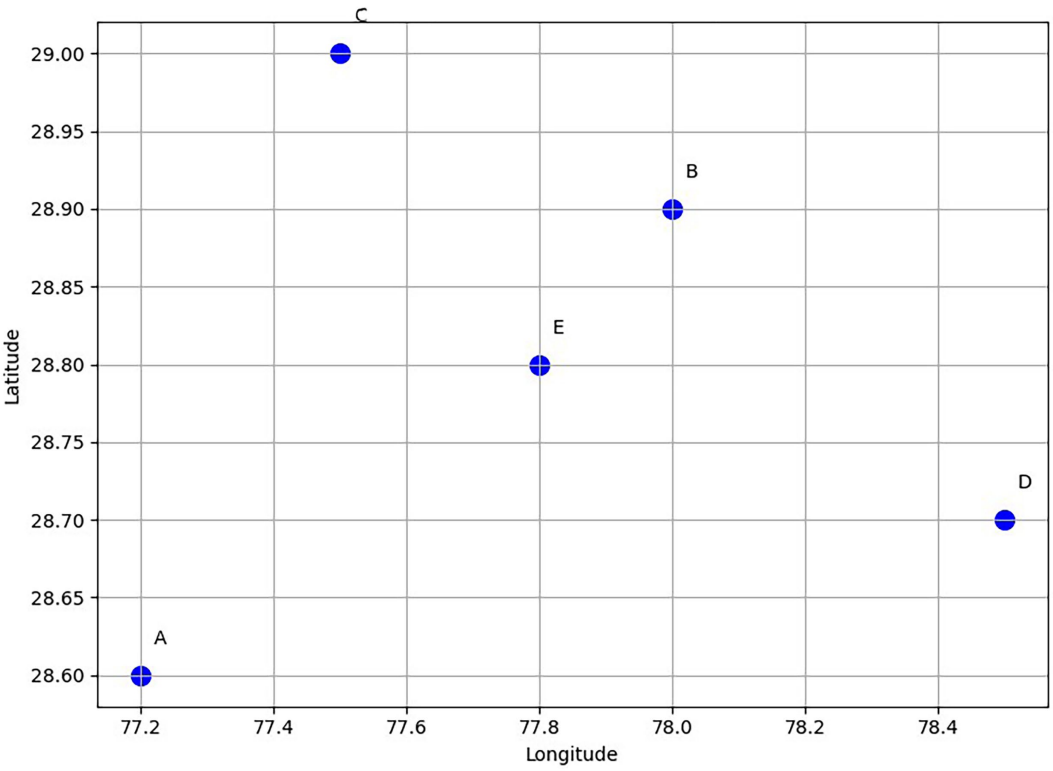


FIGURE 7
Geographical distribution of shipments.

6 Discussion

The findings of the present investigation reveal that the use of a permissioned blockchain, coupled with QR-based verification, can greatly increase transparency, traceability, and consumer confidence in agri-food supply chains. The simulated batch generation, event logging, and smart contract validation demonstrate that traceability accuracy is 97, with QR validation success rates consistently above 95, indicating a stable mapping of blockchain-registered events to real-world products. The proposed framework is scalable, with low latency (180 ms) and high throughput (180 TPS), which is better than traditional supply

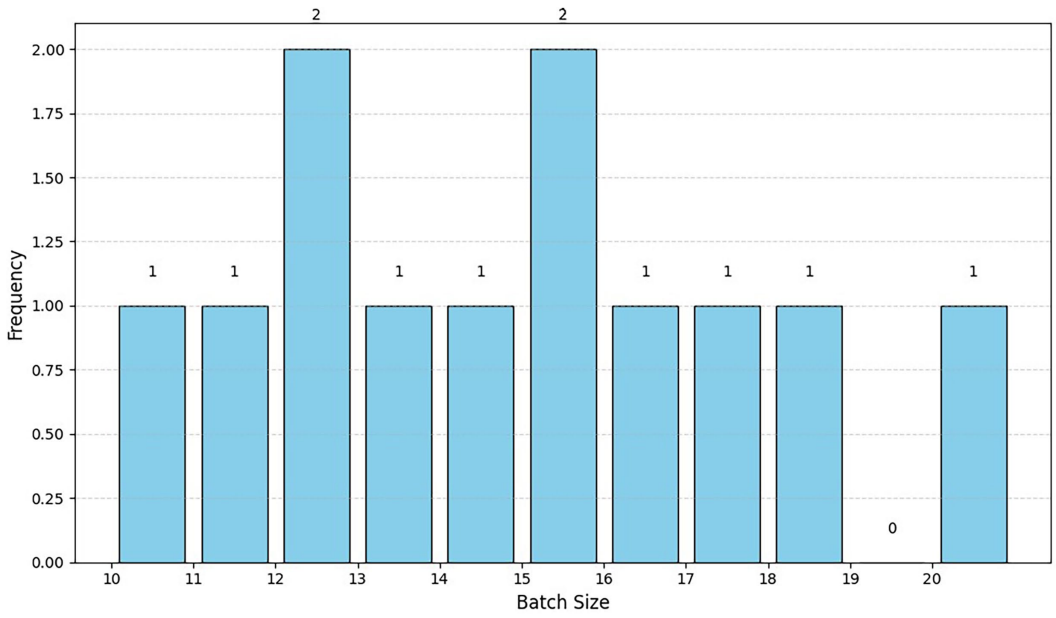


FIGURE 8
Batch size distribution.

TABLE 7 Performance comparison of proposed system with existing approaches.

Study/Metric	Throughput (TPS)	Latency (ms)	Traceability accuracy (%)	QR verification success (%)	Smart contract validation (%)
Traditional Agri Supply Systems Marchese and Tomarchio, (2022)	50	320	85	0	0
Blockchain-Based Supply Compagnucci et al. (2022)	100	250	92	85	90
Proposed Food Supply Framework	180	180	97	95	98

chains and previous blockchain-based systems. Smart contract validation at the stage of event analysis demonstrates strong adherence to supply chain rules, minimizing the risk of manipulation or fraud. The variability in supply is well controlled, as shown by batch-size distributions and traceability charts, ensuring consistency in monitoring supply throughout harvest, transport, storage, and market arrival. Moreover, comparison to the existing systems demonstrates definite improvements in automated recording of the events, resource usage, and system efficiency. These results prove that the given system will not only provide proper and safe event documentation but also consumer authentication and trust, which were lacking in the previous literature in relation to traceability and automation. All in all, the framework offers a scalable, transparent, and trust-enhancing solution for contemporary agri-food supply chains. However, adoption among small-scale farmers may face challenges, including limited digital literacy, limited infrastructure, and implementation costs. Capacity building, cost-effective technical know-how, and institutional support are essential for stakeholder engagement. However, despite the benefits offered by permissioned blockchain systems, there are also security challenges, such as insider attacks or abuse of administrative rights. By enforcing rigorous access controls, smart contract audits, and monitoring all transactions, the potential threats and risks can be reduced. Further, the proposed QR-based verification system may encounter challenges due to consumers' low digital literacy and limited smartphone and internet access. The use of intuitive user interfaces and consumer awareness campaigns is essential to support consumer adoption.

7 Conclusion and future studies

The QR-based verification and permissioned blockchain framework proposed here enhances transparency, traceability, and consumer confidence in the agri-food supply chain. The Demand Data of Top Vegetable (Kaggle) was used to generate synthetic batches and simulate traceability events to mimic real-world supply chain statuses such as harvesting, transportation, storage, and arrival at markets to evaluate the proposed framework in a simulated environment. The experiments indicate high system performance, with 97% traceability, more than 95% QR verification, and 98% smart contract verification, surpassing the performance of existing supply chain monitoring systems and several blockchain-based traceability systems. Analysis demonstrates the framework's scalability, with a throughput of 180

TPS and an average latency of 180 ms, enabling efficient, real-time monitoring of supply chain events. Blockchain-based logging and QR verification ensure secure and immutable product traceability, as well as enabling users to verify the authenticity of products themselves.

The next steps will involve incorporating IoT-based sensing technologies for automatic data gathering and minimizing human errors in recording traceability events. Machine learning techniques can support predictive demand forecasting and anomaly detection in supply chains. Also, trial implementations and field experiments will be conducted with agri-food supply chain stakeholders to test the feasibility of the proposed approach, user acceptance, and cost–benefit analysis in an actual agri-food supply chain environment. However, uptake among small farmers can be hindered by limited digital literacy, limited infrastructure, and the cost of implementation. Thus, education and training, cost-effective technical support, and policy support will be required for stakeholder engagement.

Data availability statement

The dataset used in this study is publicly available from Kaggle and can be accessed at https://www.kaggle.com/datasets/vigneshpg/demand-data-of-top-vegetables.

Author contributions

IA: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Funding

The author(s) declared that financial support was received for this work and/or its publication. This study was funded by the Deanship of Graduate Studies and Scientific Research at Jouf University under grant number: DGSSR-2026-NF-02-014.

Acknowledgments

This work was funded by the Deanship of Graduate Studies and Scientific Research at Jouf University under grant No. (DGSSR-2026-NF-02-014).

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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